Exploratory Data Analysis (EDA)

Titanic Passenger Dataset | Task 5 – Data Analyst Internship

1. Introduction

Exploratory Data Analysis (EDA) is the process of examining a dataset using descriptive statistics and visualizations to understand its structure, identify missing values and unusual observations, and discover patterns between variables. This report summarizes the EDA documented in the accompanying Jupyter Notebook for the Titanic passenger dataset.

2. Objectives

- Inspect the dataset structure, column types, and descriptive statistics.
- Measure missing values and review the distributions of numerical and categorical variables.
- Explore survival patterns by gender, passenger class, and embarkation port.
- Use histograms, bar plots, box plots, scatter plots, a pairplot, and a correlation heatmap to examine distributions and relationships.
- Summarize findings and note limitations without treating associations as proof of causation.

3. Tools and methods

- The notebook uses Python with Pandas and NumPy for data handling and descriptive analysis, and Matplotlib and Seaborn for visualizations.
- The analysis includes head(), info(), describe(), missing-value counts, value_counts(), grouped survival-rate calculations, histograms, box plots, scatter plots, a pairplot, and a numerical correlation heatmap.

4. Dataset overview and data quality

Item	Report
Passenger records	891 records are indicated by the reported gender counts (577 male + 314 female)
Main variables	survived, sex, pclass, age, fare, embarked, sibsp, parch; deck is also referenced.
Missing Values	The notebook checks missing-value counts and percentages, including age and deck.
Data types and statistics	df.info(), df.describe(), and categorical summaries are used.

5. Key findings

5.1 Survival by gender

The notebook reports a survival rate of approximately 74.2% for female passengers and 18.9% for male passengers. This is a large difference in the observed data. It describes an association and does not, by itself, establish the cause of the difference.

5.2 Survival by passenger class

The notebook reports survival rates of approximately 63.0% in first class, 47.3% in second class, and 24.2% in third class. The observed survival rate decreases from first to third class, suggesting passenger class is an important variable to investigate further.

5.3 Survival by embarkation port

Embarkation code	Reported survival rate
C — Cherbourg	55.4%
Q- Queenstown	39.0%
S — Southampton	33.7%

Passengers embarking at Cherbourg had the highest reported survival rate among the three embarkation codes. These differences may reflect other factors, including passenger class and gender, so they should not be interpreted as evidence that the port itself caused a higher survival rate.

5.4 Age and fare distributions

The notebook uses histograms to examine age and fare distributions. Its observations describe passenger ages as concentrated around younger and adult age ranges, while fares are right-skewed: many passengers paid relatively low fares and a smaller number paid substantially higher fares. The notebook also uses box plots to identify potential extreme values, particularly in fare. Exact distribution statistics should be taken from the executed notebook output.

5.5 Relationships between variables

The scatter plot of age and fare, the sibsp-versus-parch plot, and the pairplot allow multiple variables to be compared. The notebook notes that many passengers had low values for sibsp and parch, indicating that many travelled with few or no siblings/spouses or parents/children recorded aboard. These plots are useful for spotting clusters, but overlapping points and categorical variables can limit interpretation.

5.6 Correlation heatmap

The correlation heatmap summarizes linear relationships among numerical columns. Values near +1 indicate positive correlation, values near −1 indicate negative correlation, and values near 0 indicate weak linear correlation. Correlation does not establish causation. Before submitting, ensure the notebook heatmap cell ends with plt.show() so the figure is rendered.

6. Conclusion

The EDA highlights clear differences in observed Titanic survival rates by gender and passenger class, as well as variation by embarkation port. Visualizations also show why fare distributions and potential outliers should be examined carefully. Missing values, especially in fields such as age and deck, need to be documented before any downstream analysis. Overall, the analysis demonstrates how descriptive statistics and visual exploration can reveal patterns and guide more focused questions about a dataset.